

The Bin Packing Problem

An NP-complete problem and its applications in blockchain

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Why we care (motivation from blockchain)

- A block in a blockchain has a **hard limit** (block weight in Bitcoin, the gas limit in Ethereum).
- Transactions of differing “size” have to be fitted into a bounded block.
- The question “how do we lay everything out into the fewest blocks/blobs without overflowing” — that is **bin packing**.

Takeaway

Behind the everyday “the miner builds a block” sits an NP-complete problem.

Basic notions (briefly)

- **Deterministic TM** — one unambiguous next step (the model of an “ordinary” algorithm).
- **Nondeterministic TM** — several possible steps; accepts if at least one branch succeeds (“guess a certificate and check it”).
- **P** — solvable in polynomial time. **NP** — a solution is easy to **verify** in polynomial time.
- **NP-hard** — every problem in NP reduces to it. **NP-complete** — NP-hard **and** itself in NP.
- Proving NP-completeness: (1) show membership in NP; (2) reduce a known NP-complete problem to it.

Statement of the problem

- Given items of sizes $s_1, \dots, s_n$ and a bin capacity C .
- **Optimisation:** place everything into the **minimum** number of bins so that each bin's total is $\leq C$.
- **Decision version:** “are K bins enough?” — this is the NP-complete one.

$$\text{place } \{s_1, \dots, s_n\} \text{ into } \leq K \text{ bins, } \sum_{i \in B_j} s_i \leq C.$$

Proof: membership in NP

- The **certificate** is an assignment “item $\rightarrow$ bin index”.
- **Verification** in polynomial time: for each bin, sum the sizes and compare with C .
- Hence BIN PACKING $\in NP$.

Proof: NP-hardness (reduction from PARTITION)

- **PARTITION:** split numbers $a_1, \dots, a_n$ (sum $2T$) into two parts of T each.
- **Reduction:** take $s_i = a_i$, $C = T$, $K = 2$.
- $(\Rightarrow)$ A split into halves exists $\rightarrow$ two bins of $T \leq C$.
- $(\Leftarrow)$ Everything fits into 2 bins of T and the total is $2T \rightarrow$ each is loaded to exactly $T \rightarrow$ a split exists.
- So $\text{PARTITION} \leq_p \text{BIN PACKING} \rightarrow$ the problem is **NP-hard**.
- Together with membership in NP $\rightarrow$ **BIN PACKING is NP-complete.** ■

Consequence: hard even to approximate

- The same reduction: telling “2 bins” from “3” = solving PARTITION.
- Assuming $P \neq NP$, there is **no** polynomial-time algorithm with an approximation ratio better than $3/2$ (worst case).
- But asymptotically there are excellent schemes (APTAS, Karmarkar–Karp).

Applications in blockchain

- **Block construction** under the gas or size limit (alongside the knapsack problem for fees).
- **Rollups**: batching transactions into **blobs** (~ 128 KB, EIP-4844) — minimising the number of blobs = lower L2 cost.
- **Sharding**: distributing accounts and state across shards without overloading them.
- **Storage / DA**: laying out data into fixed-size sectors.

How it is solved in practice

- **Exactly** (small instances): ILP, branch-and-price (column generation).
- **Fast heuristics with guarantees:**
 - Next Fit $\leq 2 \cdot \text{OPT}$;
 - First Fit, Best Fit $\approx 1.7 \cdot \text{OPT}$;
 - **First Fit Decreasing** $\leq \frac{11}{9} \text{OPT} + \frac{6}{9}$ — sort by decreasing size and place into the first bin that fits.
- **Online** (a stream of transactions from the mempool) $\rightarrow$ greedy FF/BF; **offline** $\rightarrow$ sorting (FFD/BFD).
- In blockchain clients: order transactions by fee per unit of gas and fill the block greedily.

Conclusions

- Bin packing is a clear **NP-complete** problem with a short proof (reduction from PARTITION).
- Hard exactly and hard to approximate, but it has good practical heuristics.
- In blockchain — from block construction to rollup blobs and sharding.

Questions?